

Template Exam Solutions Document

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Evaluated criteria

- C1.** Example criterion description.
- C2.** Example criterion description.
- C3.** Example criterion description.

1 Example Uniform Investment Plan (C1)

A variable V follows a uniform model on $[8, 16]$. Use the graph and density model to answer the tasks.

Questions/Tasks.

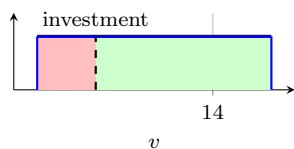
- a) Write the density function and support. [2]
 - b) Compute $P(10 \leq V \leq 14)$ and justify with area. [2]
- a) Write the density function and support.

$$f_V(v) = \begin{cases} \frac{1}{16-8} = \frac{1}{8}, & 8 \leq v \leq 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \leq v \leq 16 \quad \longrightarrow \quad f_V(v) = \frac{1}{8} \quad \longrightarrow \quad \int_8^{16} f_V(v) dv = 1$$



b) Compute $P(10 \leq V \leq 14)$ and justify with area.



$$\begin{aligned} P(10 \leq V \leq 14) &= \int_{10}^{14} \frac{1}{8} dv \\ &= \frac{1}{8}(14 - 10) = \frac{4}{8} = 0.5 \end{aligned}$$

The shaded area represents half of the support interval.



2 Example Linear Density Model (C2)

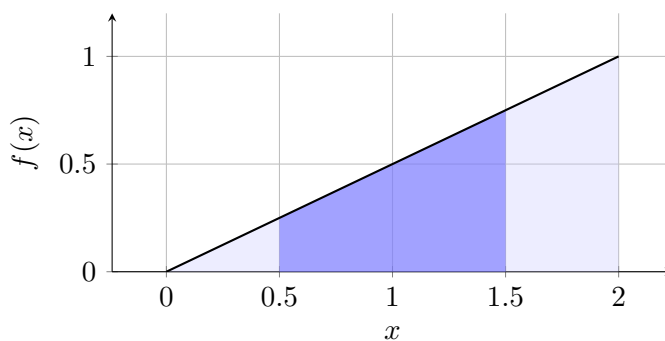
Suppose X has density $f(x) = kx$ for $0 \leq x \leq 2$.

Questions/Tasks.

- Find the value of k .
- Compute $P(0.5 \leq X \leq 1.5)$.

a) Find the value of k .

$$1 = \int_0^2 kx \, dx = k \left[\frac{x^2}{2} \right]_0^2 = 2k \quad \Rightarrow \quad k = \frac{1}{2}.$$



b) Compute $P(0.5 \leq X \leq 1.5)$.

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{1}{2}x \, dx = \left[\frac{x^2}{4} \right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$



3 Example Table and Probability Summary (C3)

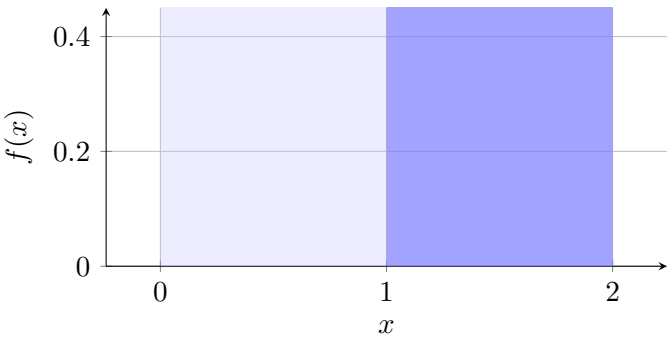
A summary table is shown for two intervals of a piecewise density.

Questions/Tasks.

- a) Complete the probability summary table.
- b) State one consistency check.

a) Complete the probability summary table.

Interval	Expression	Result
$[0, 1]$	$\int_0^1 0.4 \, dx$	0.4
$[1, 2]$	$\int_1^2 0.6 \, dx$	0.6
Total	$0.4 + 0.6$	1



b) State one consistency check.

A valid check is that all interval probabilities are nonnegative and add to 1.

